

Logistic Regression

Logistic Regression is a **supervised machine learning algorithm** mainly used for **classification problems**.

Unlike linear regression, which predicts a continuous numerical value, logistic regression predicts the **probability that an observation belongs to a particular class**.

Suppose we want to predict whether a student will Pass (1) or Fail (0) based on study hours.

The model first calculates a linear combination:

$$z = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Then it converts this value into a probability using the Sigmoid (Logistic) function:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

The probability lies between **0 and 1**.

For example:

- Probability = 0.85 → Class 1
- Probability = 0.25 → Class 0

Usually, 0.5 is used as the classification threshold.

Multinomial Logistic Regression

Used when there are **three or more classes without a natural order**.

Example: Predicting the type of transport:

- Bus
- Train
- Car

There is no meaningful order such as $\text{Bus} < \text{Train} < \text{Car}$.

The model estimates a probability for each class, and the class with the highest probability is selected.

Example:

Suppose we want to predict a student's performance based on the number of hours studied.

Student	Study Hours	Performance
A	1	Poor
B	2	Poor
C	3	Average
D	4	Average
E	5	Good
F	6	Good

Step 1: Linear equation

Logistic regression first calculates:

$$z = b_0 + b_1X$$

$$z = -5 + 1.2X$$

For a student studying **4 hours**:

$$z = -5 + (1.2)(4)$$

$$z = -0.2$$

$$P(Y = 1) = \frac{1}{1 + e^{-z}} \quad P(Y = 1) = \frac{1}{1 + e^{0.2}} \quad P(Y = 1) \approx 0.45$$

So the probability of **Pass** = 45%.

Using a threshold of 0.5:

$$P(\text{Pass}) < 0.5$$

Therefore:

Prediction = Fail

What happens if the student studies 5 hours?

$$z = -5 + (1.2)(5) = 1$$

$$P(\text{Pass}) = \frac{1}{1 + e^{-1}} \approx 0.73$$

So:

Probability of Pass = 73%

0.73 > 0.5 Prediction = Pass